Jeremy Quijano

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Summary

BS in Physics. Currently enrolled in a Computer Science Master's program at the University of South Alabama. I am interested in data analysis and data science job opportunities. I have 7 years of Python and 5 years of Machine Learning experience in a research environment from undergraduate and graduate work. I have 3 years of Data Science experience from research and fellowships. Several of my projects can be found on my GitHub.

Skills

Programming: Python 3, C++, Java, Arduino, ROS 2, Jupyter Notebook, Anaconda, Visual Studio, Visual Studio Code, PyCharm, Eclipse, Arduino IDE, GitHub

Data Science: Machine Learning, TensorFlow, Keras, PyTorch, SciKit-Learn, NumPy, Pandas, SciPy

Data Visualization/Presentations: Matplotlib, Seaborn, Plotly, Microsoft Excel and Powerpoint, Google Sheets and Slides, Apple Numbers and Keynote

General Computing: MacOS, Windows 10/11, Linux (Ubuntu), Virtual Machines (VMware Workstation Pro, VirtualBox, Parallels), Microsoft Word, Google Docs, LaTeX (Overleaf)

Electronics: PC Hardware, Arduino, Single-Board Computers, Breadboards, Basic Circuits

General Tools/Shop Equipment: Hand Tools, Power Tools, Band Saw, Lathe, Jointer, Grinder, Welding, Car Maintenance

Education

University of South Alabama (USA) — **Mobile, AL**
MS Computer Science Student — *08/2024 – Present*
- Current GPA: 4.0

The University of Alabama (UA) — **Tuscaloosa, AL**
Graduate Astrophysics Coursework — *08/2020 – 05/2023*
- 49 credit hours earned towards a PhD.

University of California, Riverside (UCR) — **Riverside, CA**
BS Physics — *09/2017 – 06/2019*

Crafton Hills College (CHC) — **Yucaipa, CA**
Transfer Credit - Physics — *08/2014 – 05/2017*

Experience

Graduate Assistant — **01/2026 – Present**
University of South Alabama, Computer Science — *Mobile, AL*
- Part-time 20 hours.
- Assisting Dr. J. Todd McDonald's research.
- Continuing master's thesis research.

ACT and Subject Tutor — **10/2024 – 08/2025**
Huntington Learning Center — *Mobile, AL*
- Part-time 10-20 hours.
- ACT Math and Science tutor.
- Math and Science subject tutor.

Esports Coach — **02/2024 – 06/2024**
Vanta Esports — *Tuscaloosa, AL*
- Virtual part-time 2-6 hours.
- Rocket League coach for middle-school and high-school teams.

Athletics Tutor — **01/2024 – 07/2024**
The University of Alabama — *Tuscaloosa, AL*
- Part-time 10 hours.
- Math and Science tutor.

Graduate Researcher **06/2021 – 06/2023**
The University of Alabama *Tuscaloosa, AL*
- Part-time 10-20 hours.
- Research under Dr. Sergei Gleyzer.
- Dark matter substructure analysis using machine learning.

Graduate Teaching Assistant **08/2020 – 05/2021**
The University of Alabama *Tuscaloosa, AL*
- Part-time 20 hours.
- Instructor for Introduction to Astronomy Lab.

Undergraduate Researcher **09/2018 – 06/2019**
University of California, Riverside *Riverside, CA*
- Unpaid part-time 10-20 hours.
- Research under Dr. George Becker.
- Senior thesis: Quasar Spectra Analysis Using Machine Learning.

Projects

Formula SAE Electric Vehicle **01/2025 – Present**
University of South Alabama
- Designing EV for 2026/2027 competition season.
- Team Lead – Computer Systems.
- Team Lead – Test Driver.

Turbulence Prediction Using Non-Linear Phase Space Analysis **08/2024 – Present**
University of South Alabama
- Master's thesis.

Formula SAE Hybrid Vehicle **10/2024 – Present**
University of South Alabama
- Prototyping pneumatic shifter for formula hybrid car.
- Team Lead – Computer Systems.
- Team Lead – Test Driver.

Strong Gravitational Lens Finding Project **06/2021 – 06/2023**
The University of Alabama
- Automated process for finding strong gravitational lenses using machine learning.

Quasar Spectra Analysis Using Machine Learning **04/2019 – 06/2019**
University of California, Riverside
- Senior thesis project.
- Machine learning classifier to study different intergalactic medium parameters.

2D Spectral Analysis and Sigma Clipping Program **09/2018 – 04/2019**
University of California, Riverside
- Research project for Dr. George Becker.
- Testing a new FITS image reduction pipeline.
- Custom sigma clipping program.

Presentations

1st Annual Shelby Hall Graduate Research Forum **03/28/2025**
University of South Alabama
- Master's Thesis Poster: Turbulence Prediction Using Nonlinear Phase Space Analysis.

Astronomy Journal Club Presentations **01/2020 – 05/2023**
The University of Alabama
- Two presentations a semester covering recent astronomy and astrophysics topics and student research.
- Presentation of UCR senior thesis: Quasar Spectra Analysis Using Machine Learning **04/02/2021**

Achievements and Awards

Davidson Family Reboot Into Computing Endowed Graduate Scholarship **08/2024**
University of South Alabama
- $1000 scholarship for the 2024-2025 academic school year.

LSSTC Data Science Fellowship **10/2021 – 07/2023**
The University of Alabama
- Data Science fellowship sponsored by Northwestern University.
- Data Science workshops for graduate Astronomy and Astrophysics students.

Southern Regional Education Board Fellowship **08/2021 – 06/2023**
The University of Alabama
- SREB State Doctoral Scholars Program member
- Provides support for minorities pursuing PhD's and continuing in academia.

2017 Southern California Edison STEM Scholarship **06/2017**
Crafton Hills College
- $2500 scholarship for achieving academic excellence and continue pursuit of a STEM degree.

Extracurriculars

Society of Automotive Engineers (SAE)
University of South Alabama
- Officer – Operations Manager 11/2025 – Present
- Officer – Co-Operations Manager 10/2025 – 11/2025
- General Member 10/2024 – 10/2025

Jag Racing
University of South Alabama
- Formula SAE Team at USA.
- Team Lead – Computer Systems 08/2025 – Present
- Team Lead – Test Driver 08/2025 – Present
- Electrical Systems 10/2024 – Present

Esports
- USA Rocket League Player and Coach 08/2024 – 12/2024.
- UA Rocket League Player and Coach 08/2020 – 06/2022.
- UCR Rocket League Player and Coach 09/2017 – 06/2020.

U13+ Youth Recreational Soccer Coach **03/2024 – 05/2024**
Tuscaloosa United Soccer Club *Tuscaloosa, AL*
- Head Coach for U13+ soccer team.

Google Summer of Code Mentor **06/2022 – 08/2022**
The University of Alabama
- Machine Learning mentor for UA DeepLense Project.

Physics Club **08/2016 – 05/2017**
Crafton Hills College
- Local STEM outreach.